

GOVERNMENT OF ANDHRA PRADESH
DEPARTMENT OF MINES & GEOLOGY ::MANGALAGIRI

Memo No. INC04-16/21/2026-IT SEC - DMG Date:30-07-2026.

Sub:- Mines & Minerals - EY-RTGS PMU - **Smart Granite Block Measuring and Tax Assessment Hackathon** to develop an AI-based solution for automated granite block measurement, seigniorage assessment, and integration with OMEPS 2.0. – Nomination of Jury from the Department - Reg.

Ref:- E-mail received from Swagath Biswas EY-RTGS PMU team dt:25-06-2026.

Adverting to the subject and reference cited :

The EY-RTGS PMU has informed about the above **Smart Granite Block Measuring and Tax Assessment Hackathon** to develop an AI-based solution for automated granite block measurement, seigniorage assessment, and integration with OMEPS 2.0, and requested the Department requested to nominate jury members and to accord acceptance for bearing and releasing the below said prize amounts to the winning teams.

Regarding Granite Mineral, currently undergoes manual block measurement by DMGO officers during field inspections. Officers physically measure the length, breadth, and height of granite blocks, compute volume manually, and enter the details into the mining administration system for seigniorage fee assessment on a per Metric Tonne (MT) basis.

The existing process presents several operational and governance challenges:

- Manual measurement is subjective, leading to inconsistent seigniorage assessments across quarries.
- Potential revenue leakage due to discrepancies between field measurements and actual block volumes.
- No audit trail linking block-level measurement to dispatch and weighbridge records.
- High resource and cost burden of repeated field inspections across dispersed quarry sites.

To address these issues, the Department proposes a technology-driven solution through the RTGS AI Hackathon Initiative.

Expected Solution:

Participants are expected to design and develop an AI/Computer Vision-based solution capable of

1. Granite Block Volume Estimation

- Estimate granite block dimensions and volume using standardized mobile-captured images.
- Support accurate measurement from field photographs with minimal manual intervention.

2. Automated Seigniorage Classification

- Automatically classify granite blocks into applicable seigniorage fee categories based on block dimensions and classification rules.

3. Mobile Application for Field Capture

- Develop a mobile application for lessees/quarry operators with the following capabilities.
- GPS-tagged image capture
- Time stamped records
- Block ID capture
- Quarry details capture
- Standardized image acquisition process

4. System Integration

- Integrate with OMEPS 2.0 for cross-validation of
- AI-computed block volumes
- Weighbridge measurements
- Dispatch returns

5. Audit and Traceability

- Enable digital traceability from block measurement to transportation and dispatch lifecycle.

In view of the above, the following officers are nominated as Jury for Conducting the participant orientation session, and evaluating the final solution demonstrations /presentations shortlisting.

Departmental Jury Nomination for the final selection of winners

Sl. No.	Name	Designation
1	Sri D. Srinivasa Rao	Additional Director
2	Sri G. Srinivasa Chowdary	DD(Mines)

3	Sri A. Srinivasa Kumar	DDM&G (O/o. DM&GO, Vijayawada)
4	Sri V. Ravichand	DDM&G (O/o. DM&G, Mangalagiri)
5	Sri T. Bala Subramanyam	ADM&G (O/o. DM&G, Mangalagiri)

In view of the above, the above Jury members are hereby requested to make it convenient to participate in the both orientation session and the final hackathon demo/presentation evaluation as per the scheduled communicated by the IT Section.

Encl: (as above)

BALA CHANDRA SEKHAR W
DIRECTOR OF MINES AND GEOLOGY

To
The above Officers.